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Assignment 11: Network monitoring with Wireshark

https://github.com/elovric13/eutech/tree/network-monitoring-with-wireshark-elovric13

1. Install Wireshark on a workstation or VM with network access

Wireshark was installed on an Ubuntu VM. During installation, non-superuser packet capture was enabled, and the current user was added to the wireshark group to allow capturing without running as root.

The active network interface (ens33) was selected from the capture interface list and used for all captures in this report.

2. Capture network traffic for at least 10 minutes

Traffic was captured for over 10 minutes.

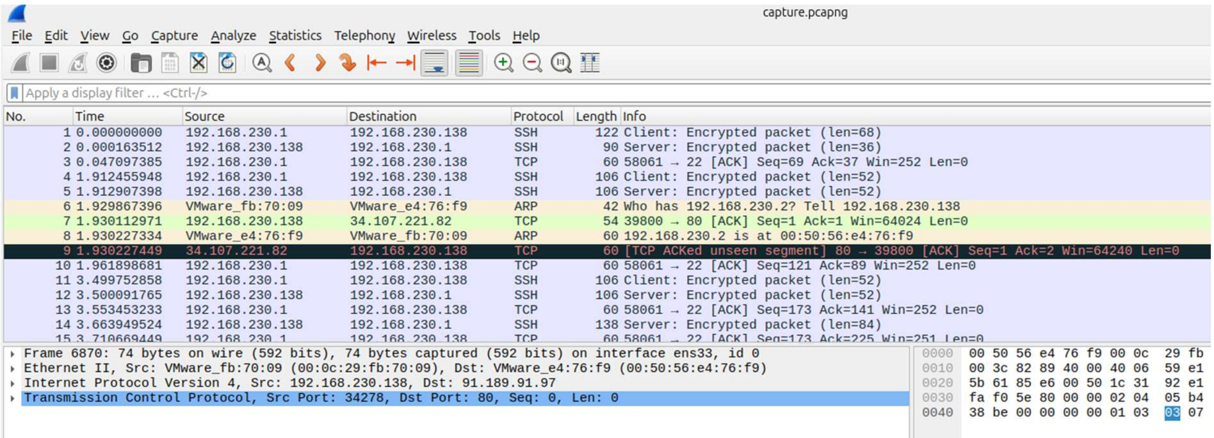

Figure 1:Wireshark capture

3. Filter and analyze packets related to HTTP/HTTPS, SSH, and DNS

- **HTTP traffic**

The http filter was applied to isolate HTTP requests and responses. The capture showed multiple GET requests from the local machine (192.168.230.138) to various destination IPs, with corresponding 200 OK, 301 Moved Permanently, and 204 No Content responses.

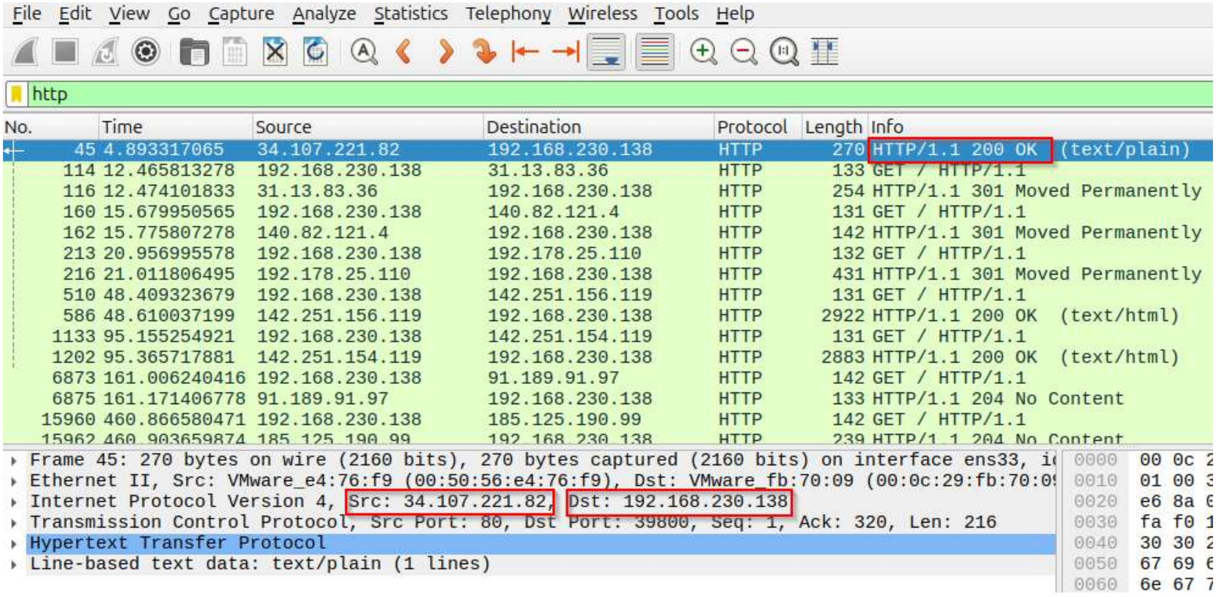

Figure 2: HTTP filter

- HTTPS traffic

The tls filter was applied to isolate encrypted HTTPS traffic. The capture clearly showed a TLS 1.3 handshake, including a Client Hello packet with the SNI (Server Name Indication) field revealing the requested domain, followed by a Server Hello response from the destination server.

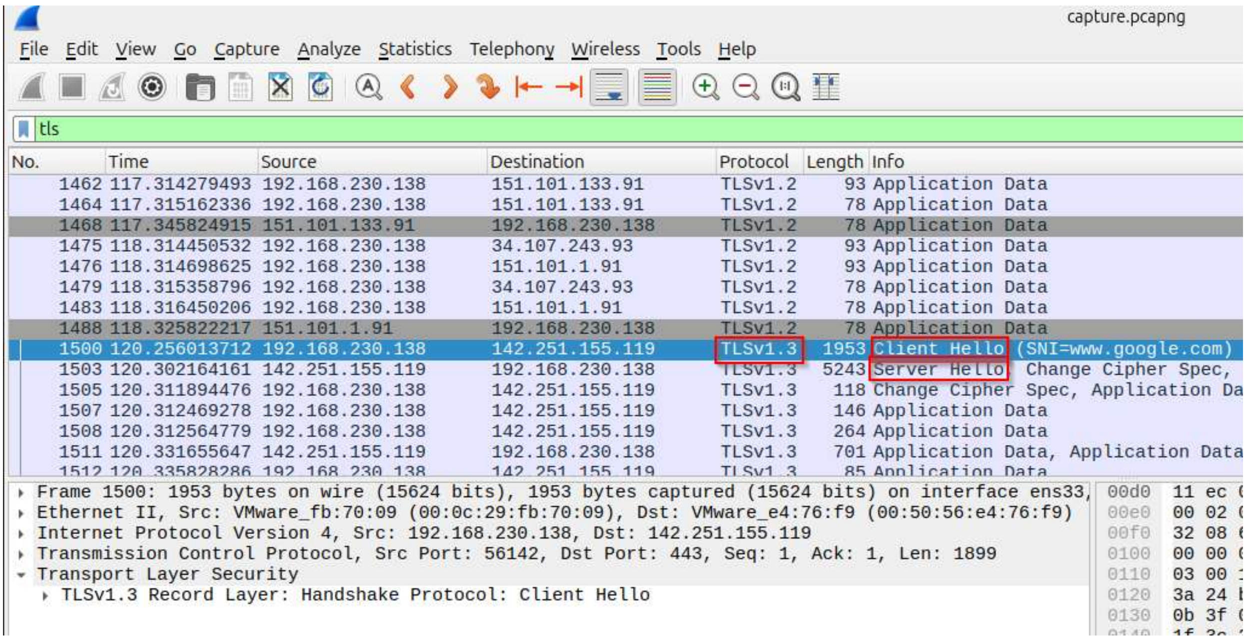

Figure 3: TLS handshake

- DNS traffic

The dns filter was applied to isolate domain name resolution traffic. The capture showed standard DNS queries and responses, confirming domain names being resolved to their corresponding IP addresses by the local DNS resolver (192.168.230.2).

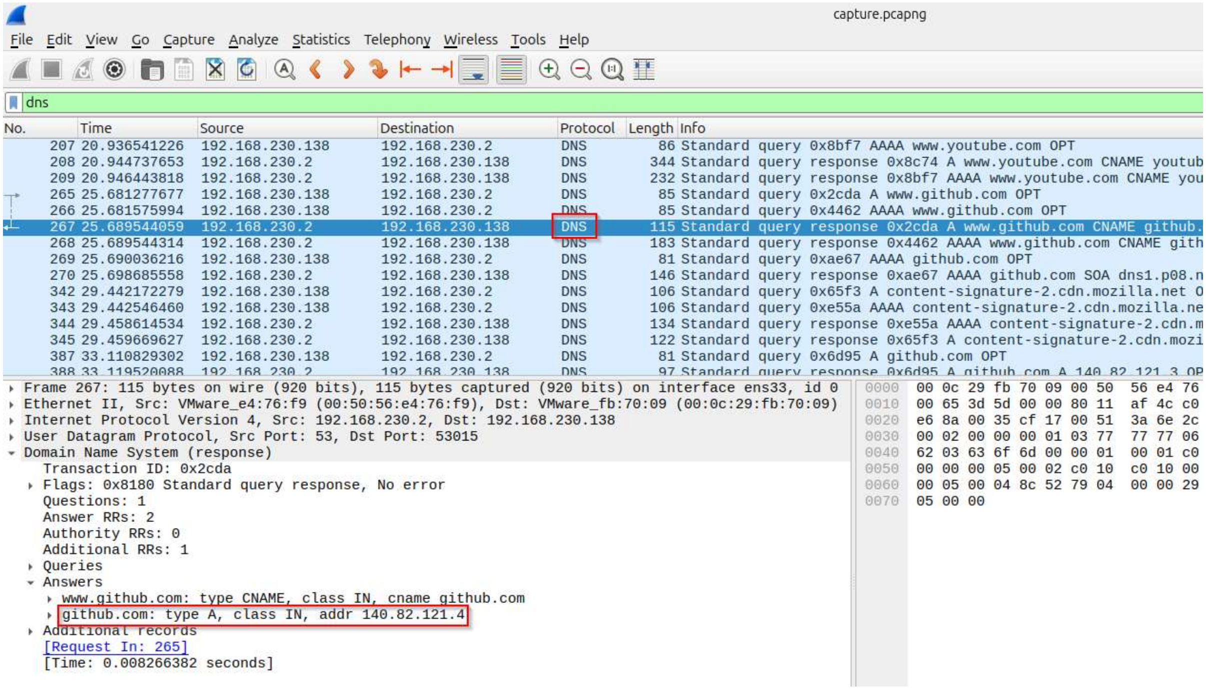

Figure 4: DNS query response

- SSH traffic

The ssh filter was applied to isolate SSH traffic from a local SSH connection to localhost. As expected with SSH, no readable content was visible since all packets were encrypted, shown as 'Encrypted Packet' in both the client-to-server and server-to-client directions.

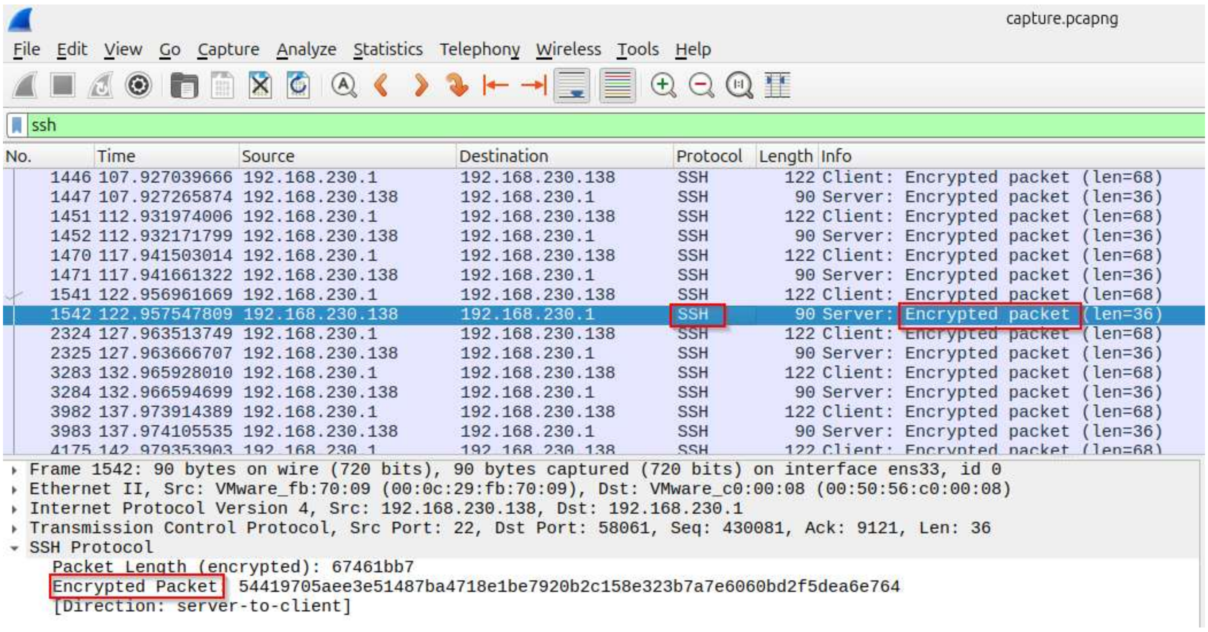

Figure 5: SSH traffic

4. Identify IP addresses, protocols, and possible suspicious access attempts

Protocol	Local IP	Remote IP	Purpose
HTTP	192.168.230.138	34.107.221.82	Plaintext web request
TLS/HTTPS	192.168.230.138	142.251.155.119	Encrypted web request (google.com)
DNS	192.168.230.138	192.168.230.2	Domain name resolution
SSH	192.168.230.138	192.168.230.1	Encrypted remote shell session

- Checking for suspicious access attempts

The filter tcp.flags.syn == 1 && tcp.flags.ack == 0 was used to isolate new outbound connection attempts. All SYN packets were directed to standard ports (80, 443) across expected destination IPs, with no repeated connections to a single IP and no unusual ports. No suspicious activity was identified.

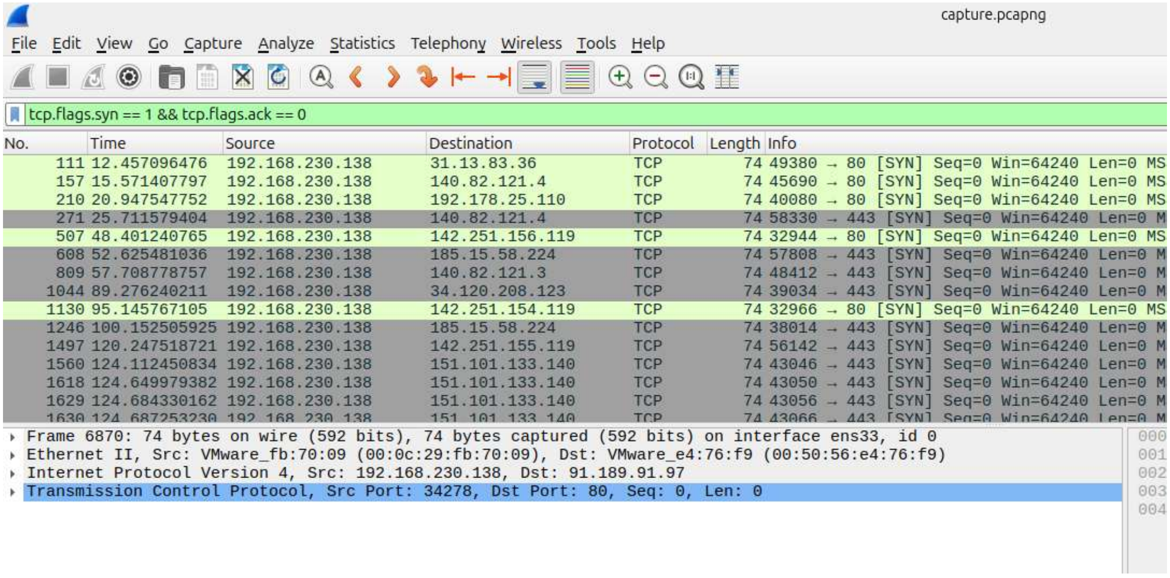

Figure 6: SYN filter

This task demonstrated the use of Wireshark to capture and analyze live network traffic. By applying protocol-specific filters, it was possible to clearly isolate and examine HTTP, HTTPS, DNS, and SSH traffic, identify the IP addresses involved in each type of communication, and verify that no suspicious connection patterns were present in the captured traffic.